

Taehoon Kim

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Research Interests

My research aims to develop robust perception for real-world, safety-critical environments through multimodal representation learning with computer vision and vision–language models. I am particularly interested in spatial intelligence and autonomous-driving perception via multi-sensor fusion (RGB, LiDAR, Radar). I also study industrial and multimodal anomaly detection via distillation-based teacher–student frameworks.

Keywords: Computer Vision, Spatial Intelligence, Multi-Sensor Fusion, Anomaly Detection, Vision-Language Model

Education

Kyonggi University

M.S. in Computer Science

- Advisor: Junho Ahn
- GPA: 4.5 / 4.5

Sep. 2025 – Present

Suwon, Korea

Kyonggi University

B.S. in Department of Restaurant and Culinary

B.S. in Department of Artificial Intelligence – *double major*

- CGPA: 4.3 / 4.5 – Summa Cum Laude, AI major CGPA: 4.3 / 4.5

Mar. 2019 – Aug. 2025

Seoul, Korea

Suwon, Korea

Experiences

K-SW Square, Purdue University

Visiting Research Intern

Advisor: Eric T. Matson & Anthony Smith

Sep. 2023 – Dec. 2023

West Lafayette, IN, USA

AI Vision Lab, Kyonggi University

Undergraduate Research Assistant

Advisor: Prof. Junho Ahn

May. 2024 – Aug. 2025

Suwon, Korean

Military–Academia Joint AI Education Program, ROK DCC

Teaching Assistant & Project Mentor

2024 – 2025

Suwon, Korean

National Center of Excellence in Software, Kyonggi University

Graduate Teaching Assistant

Sep. 2025 – Present

Suwon, Korean

Publications

International Journal

- [1] **Taehoon Kim**, Sehun Lee, Namgi Kim, Youjin Seo, “XSANet: Industrial X-ray Anomaly Detection via Spatial Transformer and Attention-Enhanced Teacher-Student Framework,” *Image and Vision Computing (IMAVIS)*, Elsevier, *under review*
- [2] **Taehoon Kim**, Sehun Lee, Junho Ahn, “Zero-Shot Vision-Based Robust 3D Map Reconstruction and Obstacle Detection in Geometry-Deficient Room-Scale Environments,” *Computers, Materials & Continua*, vol. 86, no. 2, pp. 1–30, Dec. 2025.
- [3] Sehun Lee, **Taehoon Kim**, Sookyun Kim, Junho Ahn, Namgi Kim, “Unsupervised Structural Defect Classification via Real-Time and Noise-Robust Method in Smartphone Small Modules,” *Electronics (MDPI)*, vol. 14, no. 17, art. no. 3455, Aug. 2025.
- [4] Sehun Lee, **Taehoon Kim**, Junho Ahn, “Zero-Shot Based Spatial AI Algorithm for Up-to-Date 3D Vision Map Generations in Highly Complex Indoor Environments,” *Computers, Materials & Continua*, vol. 84, no. 2, pp. 3623–3648, Jul. 2025.

International Conference

- [1] Doyoung Koh, **Taehoon Kim**, Sehun Lee, Namgi Kim, “Weighted Knowledge Distillation with DisturbLabel for Robust Learning,” 2026 17th International Conference on Ubiquitous and Future Networks (ICUFN), Milan, Italy, Jul. 2026.

Domestic Journal

- [1] **Taehoon Kim**, Jiyoung Na, Jiwon Yoon, Sehun Lee, Junho Ahn, “Multiple Camera-Based Real-Time Long Queue Vision

Algorithm for Public Safety and Efficiency,” Journal of the Korea Society of Computer and Information (JKSCI), vol. 29, no. 10, pp. 47–57, Oct. 2024. DOI: 10.9708/jksci.2024.29.10.047

Domestic Conference

- [1] **Taehoon Kim**, Sehun Lee, Junho Ahn, “Industrial X-ray UniNet: A Robust Spatial Transformation Architecture for X-ray Anomaly Detection in Smartphone Actuator Modules,” Proceedings of the 2025 KAD Winter Conference (**Outstanding Paper Award**)
- [2] Sehun Lee, **Taehoon Kim**, Junho Ahn, “Genetic Neural Networks: Efficient Training Initialization Strategy via Weight Recombination,” Proceedings of the 2025 KAD Winter Conference (**Best Paper Award**)
- [3] **Taehoon Kim**, Sehun Lee, SeungBeom Lee, Junho Ahn, “0-VISTA: Zero-Shot Spatial AI for 3D Mapping and Detection in Complex Indoor Environments,” Proceedings of the 2025 KAD Spring Conference
- [4] Sehun Lee, **Taehoon Kim**, Doyeon Kim, Junho Ahn, “Latent Hotspot Crossover for Disentangled and Interpretable Representation Learning in Variational Autoencoders,” Proceedings of the 2025 KAD Spring Conference (**Outstanding Paper Award**)
- [5] Doyeon Kim, SeungBeom Lee, **Taehoon Kim**, Junho Ahn, “Detection of Movable Object Layout Changes in Shared-Use Facilities,” Proceedings of the 2025 KAD Spring Conference
- [6] **Taehoon Kim**, Jiyoung Na, Sehun Lee, Jiwon Yoon, Junho Ahn, “Deep Learning Vision-Based Line Recognition Algorithm Suggestion for Real-Time Intercity Bus Passenger Queue Efficiency,” Proceedings of the 2024 KSII Spring Conference
- [7] Yechan Lee, **Taehoon Kim**, Jiyoung Na, Jiwon Yoon, Sehun Lee, Hanyong Lee, “Analysis of Autonomous Driving Algorithms for Road Driving of Delivery Robots,” Proceedings of the 2023 KIIT Summer Conference

Patent

Deep Learning-Based Queue Recognition System	Oct. 2024
Korea Patent Application	<i>No. 10-2024-0144283</i>

Projects

Pedestrian Safety System for Large-Scale Buildings Using 3D Congestion Mapping with Collaborative Robots and Vision AI for Disaster Response NRF	Sep. 2025 – Aug. 2026
Acoustic-Based Real-Time Drone Localization & Tracking AI System ITSA & ROK DCC	Jul. 2025 – Dec. 2025
Improvement of Detect Detection and Classification Capabilities in SUB2 ASS'Y GAP Inspection im co., ltd.	Nov. 2024 – Feb. 2025
Real-Time Multimodal Vision-Integrated Military Fitness Assessment ITSA & ROK DCC	Sep. 2024 – Dec. 2024
Improvement of Detect Detection and Classification Capabilities in X-RAY Ball Inspection im co., ltd	Jul. 2024 – Aug. 2024
Enhancing Culinary Experience: Real-Time Interactive Cooking Assistance Utilizing Mixed Reality Purdue University	Sep. 2023 – Dec. 2023

Awards

Best Paper Award, 2025 Korea Data Science Winter Conference	Dec. 2025
Outstanding Paper Award, 2025 Korea Data Science Winter Conference	Dec. 2025

Outstanding Paper Award , 2025 Korea Data Science Spring Conference	Jun. 2025
Merit Award , Kyonggi University SW Imagination Company Project Competition	Nov. 2024
Gold Award , Kyonggi University Advanced Capstone Design Project Competition	Jun. 2024
Gold Award , Kyonggi University Foundational Capstone Design Project Competition	Jun. 2023
Grand Award , Kyonggi University GamsungSW IT Competency Competition	Dec. 2019

Honors

Letter of Appreciation , ROK Defense Communication Command · Award for outstanding contribution in AI development and mentoring	Apr. 2026
AI & SW Talent Scholarship , Noh Youngbaek Scholarship Foundation · Award for excellence in the AI & Software field	Nov. 2025
Faculty Council Honors Scholarship , Kyonggi University · Award for outstanding academic achievement	Jun. 2025
Highest Honors , Kyonggi University · Graduate 1st in Dept. of Restaurant & Culinary	2025
Semester High Honors , Kyonggi University · Award for perfect GPA of 4.5 / 4.5	2024
Semester High Honors , Kyonggi University · Award for Academic Excellence (1st)	2019, 2022 · 2023 (1st Sem), 2024

Technical Skills

Programming languages

Python, C, Java, R

Deep Learning Framework

Pytorch, Tensorflow, Keras, Numpy, Cuda

Language Skills

TOEIC – Total Score 920	Aug. 2024
TOEIC Writing – Advanced Mid (AM)	Apr. 2024
TOEIC Speaking – Intermediate High (IH)	Apr. 2024
Official English Presentation as University Representative Kyonggi University National Center of Excellence in Software · Presented program highlights and achievements to a CHED delegation, Philippine President's office	Jul. 2024 news article
English Interpretation & Translation Volunteer Purdue Military Research Institute (PMRI) & IITP · Interpreted AI lectures and supported communication for ROK ROTC officers	Oct. 2023